Centered-encoding parametrization for the MMS sweep

Working note (not yet merged into `article.tex`)

2026-05-28

1 Motivation

The stabilized FE solver in this codebase is dimensional. Its inputs are $(\rho, \mu, \sigma, U, L, \boldsymbol{f})$ in a chosen system of units. The Method of Manufactured Solutions (MMS) test harness, however, organises runs by the *dimensionless* regime (Re, Da, α_0), and is free to choose any consistent dimensional encoding of each dimensionless cell. Here α_0 is the infimum of the porosity field $\alpha(x)$, and $\alpha_\infty = 1$ is its supremum (a convention shared with `article.tex`).

Until recently, the harness fixed

$$U = 1, \qquad L = 1, \tag{1}$$

and derived ν, σ from the dimensionless definitions used in `article.tex`:

$$\nu \;=\; \frac{U\,L}{Re} \;=\; \frac{1}{Re}, \qquad \sigma \;=\; \frac{Da\ \alpha_\infty\,\nu}{L^2} \;=\; \frac{Da\ \alpha_\infty}{Re}. \tag{2}$$

For the sweep grid $Re \in \{10^{-6}, 1, 10^{6}\}$, $Da \in \{10^{-6}, 1, 10^{6}\}$, $\alpha_0 \in \{1, 0.5, 0.05\}$ (with $\alpha_\infty = 1$), this encoding produces values such as $\nu = 10^6$ and $\sigma = 10^{12}$ for the cell $(Re = 10^{-6}, Da = 10^6, \alpha_0 = 1)$. Twelve orders of magnitude separate the smallest from the largest dimensional coefficient that the solver sees, *all on the same side of unity.*

While the dimensionless physics is unchanged, this one-sided spread breaks numerical components of the stabilized solver that are not scale-invariant once round-off enters:

- the OSGS staggered fixed-point's L^2 projection π_h of the strong residual evaluates $\sigma\,\boldsymbol{u}_h$ and $\nu\Delta\boldsymbol{u}_h$ at near-precision-floor $\boldsymbol{u}_h$ when $\boldsymbol{u}_h \sim 1/\sigma$, amplifying floating-point noise into a non-contractive π_h update sequence;
- the assembled Jacobian's row magnitudes span 12 orders, which the direct factorisation absorbs but the Newton merit function and line-search tolerances do not;
- the MMS forcing oracle, computed by symbolic differentiation of the analytic exact solution, mixes $\sigma\,\boldsymbol{u}_{\mathrm{ex}}$ at magnitude 10^{12} with sub-dominant terms at 10^6, losing six digits of relative precision in the sub-dominant contributions.

2 The centered encoding

For a fixed dimensionless cell (Re, Da, α_∞), the harness has *two* free dimensional parameters (U, L). The freedom follows from the constraint pair (matching `article.tex` eq. 3.4):

$$Re = \frac{U\,L}{\nu}, \qquad\qquad Da = \frac{\sigma\,L^2}{\alpha_\infty\,\nu}. \tag{3}$$

Eliminating ν, σ:

$$\nu\,\sigma \;=\; \frac{\alpha_\infty\ Da\ U^2}{Re^2}, \qquad \frac{\nu}{\sigma} \;=\; \frac{L^2}{\alpha_\infty\ Da}. \tag{4}$$

So U controls the geometric mean and L controls the spread.

Choice. The *centered encoding* chooses U so that $\sqrt{\nu\,\sigma} = 1$, leaving $L = 1$ because the manufactured solution shape $\sin(\pi x)$ is hard-coded for a unit domain (see `Paper2DMMS` in `src/problems/mms_paper_2d.jl`):

$$\boxed{U \;=\; \frac{Re}{\sqrt{\alpha_\infty\ Da}}, \qquad L \;=\; 1.} \tag{5}$$

The resulting dimensional coefficients are

$$\nu \;=\; \frac{1}{\sqrt{\alpha_\infty\ Da}}, \qquad \sigma \;=\; \sqrt{\alpha_\infty\ Da}. \tag{6}$$

One checks immediately that $\nu\,\sigma = 1$ (centering condition) and $\nu/\sigma = 1/(\alpha_\infty\ Da)$ (intrinsic spread). Note that ν and σ no longer depend on Re — only U absorbs the Re variation. The spread $\sigma/\nu = \alpha_\infty\ Da$ is *intrinsic* to the cell and cannot be compressed by any choice of (U, L); centering only shifts it to be symmetric about unity.

What this does *not* change. The dimensionless equation (and therefore the dimensionless solution $\boldsymbol{u}^*, p^*$) is identical between the default and centered encodings; this is a strict reparametrization. The discrete velocity *degrees of freedom* are $\boldsymbol{u}_h^{\text{node}} \approx U\,\boldsymbol{u}^{*,\text{node}}$, so they take on different absolute magnitudes between encodings, but their dimensionless counterparts $\boldsymbol{u}^{*,\text{node}}$ are unchanged. The H^1-seminorm error $|\boldsymbol{u}_h - \boldsymbol{u}_{\text{ex}}|_{H^1(\Omega)}\,/\,(U/L)$ is regime-invariant (verified empirically: agreement to $<1\%$ across encodings on the easy regime C2).

3 Per-cell parameter table

The harness sweeps $Re \in \{10^{-6}, 1, 10^6\}$, $Da \in \{10^{-6}, 1, 10^6\}$, $\alpha_0 \in \{1, 0.5, 0.05\}$, with $\alpha_\infty = 1$ in all cells. Three cells are skipped by config ($(Re, Da, \alpha_0) = (10^6, \star, 0.05)$, three triples), leaving 24 that are exercised by the sweep. Table **??** reports the dimensional values for both encodings. Because the dimensional σ in the code path (`sigma_c = Da * alpha_infty * nu_calc / L^2`) uses $\alpha_\infty = 1$ throughout, the σ columns are independent of α_0 — they are determined purely by (Re, Da).

Maximum dimensional coefficient. The worst cells under the default encoding (C7, C16, C23) hit $\sigma = 10^{12}$. Under the centered encoding the worst is $\sigma = 10^3$ (any cell with $Da = 10^6$). The maximum dimensional coefficient drops by nine orders of magnitude across the sweep.

4 Verification evidence

Two single-cell probes (`test/extended/ManufacturedSolutions/diagnostics/jacobian_equilibration_osgs_` validated the reparametrization before the full rerun. Note that the probe used a slightly different centering formula $U = Re\,/\sqrt{\alpha_0\,Da}$ (with $\alpha_0 = 0.5$ for C16), so $U = 1.41 \cdot 10^{-9}$ and $\sigma = 1.41 \cdot 10^3$ in the probe; the harness rerun uses $U = Re\,/\sqrt{\alpha_\infty\,Da}$ with $\alpha_\infty = 1$, giving $U = 10^{-9}$ and $\sigma = 10^3$ exactly.

#	Dimensionless inputs			Default ($U = L = 1$)			Centered		
	Re	*Da*	α_0	U	ν	σ	U	ν	σ
C1	10^{-6}	10^{-6}	1.00	1	10^{6}	1	10^{-3}	10^{3}	10^{-3}
C2	1	10^{-6}	1.00	1	1	10^{-6}	10^{3}	10^{3}	10^{-3}
C3	10^{6}	10^{-6}	1.00	1	10^{-6}	10^{-12}	10^{9}	10^{3}	10^{-3}
C4	10^{-6}	1	1.00	1	10^{6}	10^{6}	10^{-6}	1	1
C5	1	1	1.00	1	1	1	1	1	1
C6	10^{6}	1	1.00	1	10^{-6}	10^{-6}	10^{6}	1	1
C7	10^{-6}	10^{6}	1.00	1	10^{6}	10^{12}	10^{-9}	10^{-3}	10^{3}
C8	1	10^{6}	1.00	1	1	10^{6}	10^{-3}	10^{-3}	10^{3}
C9	10^{6}	10^{6}	1.00	1	10^{-6}	1	10^{3}	10^{-3}	10^{3}
C10	10^{-6}	10^{-6}	0.50	1	10^{6}	1	10^{-3}	10^{3}	10^{-3}
C11	1	10^{-6}	0.50	1	1	10^{-6}	10^{3}	10^{3}	10^{-3}
C12	10^{6}	10^{-6}	0.50	1	10^{-6}	10^{-12}	10^{9}	10^{3}	10^{-3}
C13	10^{-6}	1	0.50	1	10^{6}	10^{6}	10^{-6}	1	1
C14	1	1	0.50	1	1	1	1	1	1
C15	10^{6}	1	0.50	1	10^{-6}	10^{-6}	10^{6}	1	1
C16	10^{-6}	10^{6}	0.50	1	10^{6}	10^{12}	10^{-9}	10^{-3}	10^{3}
C17	1	10^{6}	0.50	1	1	10^{6}	10^{-3}	10^{-3}	10^{3}
C18	10^{6}	10^{6}	0.50	1	10^{-6}	1	10^{3}	10^{-3}	10^{3}
C19	10^{-6}	10^{-6}	0.05	1	10^{6}	1	10^{-3}	10^{3}	10^{-3}
C20	1	10^{-6}	0.05	1	1	10^{-6}	10^{3}	10^{3}	10^{-3}
(skipped: $Re = 10^6, Da = 10^{-6}, \alpha_0 = 0.05$)									
C21	10^{-6}	1	0.05	1	10^{6}	10^{6}	10^{-6}	1	1
C22	1	1	0.05	1	1	1	1	1	1
(skipped: $Re = 10^6, Da = 1, \alpha_0 = 0.05$)									
C23	10^{-6}	10^{6}	0.05	1	10^{6}	10^{12}	10^{-9}	10^{-3}	10^{3}
C24	1	10^{6}	0.05	1	1	10^{6}	10^{-3}	10^{-3}	10^{3}
(skipped: $Re = 10^6, Da = 10^6, \alpha_0 = 0.05$)									

Table 1: Dimensional parameter values for each dimensionless cell in the `phase1_quad_k1` sweep, under the default encoding ($U = L = 1$) and the centered encoding ($U = Re / \sqrt{\alpha_\infty Da}$, $L = 1$), with $\alpha_\infty = 1$ as run by the test harness (`run_test.jl:76`). The dimensionless physics is identical between the two columns; only the dimensional representation differs. Under the centered encoding, ν and σ depend only on Da, never on Re or α_0.

C16 OSGS (high-σ, the canonical fold case).

	Default	Centered (probe)
ν	10^6	$1.41 \cdot 10^{-3}$
σ	10^{12}	$1.41 \cdot 10^{3}$
OSGS staggered iters	25 (budget)	0 (immediate)
π_u-drift	$7.6 \cdot 10^8$	$1.03 \cdot 10^{-6}$
`overall_converged`	`false`	`true`
Initial residual	$1.27 \cdot 10^{-3}$	$3.08 \cdot 10^{-10}$
$L^2(\boldsymbol{u})$	$1.07 \cdot 10^{-3}$	$4.72 \cdot 10^{-4}$
$H^1(\boldsymbol{u})$	$2.45 \cdot 10^{-1}$	$1.04 \cdot 10^{-1}$

The OSGS staggered fixed-point's projection drift drops by *fourteen orders of magnitude*. Same physics; the difference is the floating-point conditioning of the strong-residual projection.

C2 (easy regime; consistency check). With $Re = 1$, $Da = 10^{-6}$, $\alpha_0 = 1$, both encodings should give the same answer up to round-off.

	Default ($U = 1$)	Centered ($U = 10^3$)	
$H^1(\boldsymbol{u})$	$3.612 \cdot 10^{-2}$	$3.564 \cdot 10^{-2}$	(0.7% diff)
$L^2(\boldsymbol{u})$	$5.92 \cdot 10^{-4}$	$1.18 \cdot 10^{-4}$	(different noise floor)

The H^1 semi-norm error agrees to under 1% between encodings, confirming that the centered encoding represents the same discrete physics. The L^2 values differ by a constant noise-floor factor: in both runs the absolute residual at convergence is $O(10^{-9})$, and that absolute noise is divided by $U_c = U$ when forming the normalized error — a 1000× different denominator produces the apparent 5× discrepancy without any discretisation disagreement.

5 Implementation

The implementation is a single-line change to the MMS harness; no modification to the dimensional solver code:

```
# test/extended/ManufacturedSolutions/run_test.jl, ~line 523

# Before (default encoding):
U_amp = 1.0
L     = 1.0

# After (centered encoding):
U_amp = Re / sqrt(alpha_infty * Da)
L     = 1.0
```

The remainder of the harness pipeline (`build_mms_formulation`, `Paper2DMMS`, `evaluate_exactness_diagnosti` `get_characteristic_scales`) already threads U and L through correctly: the MMS exact velocity is multiplied by U, the manufactured forcing is computed analytically from that scaled solution, and error norms divide out $U_c = U$ to report dimensionless quantities. No tooling surgery is required.

6 What is deliberately *not* addressed here

- **The L degree of freedom.** `Paper2DMMS` hard-codes the spatial shape $\sin(\pi x_1)\sin(\pi x_2)$ etc., which is one half-period on the unit square but high-frequency garbage on $L \neq 1$. Allowing L to vary would require generalising the shape to $\sin(\pi x/L)$ and re-deriving the analytic gradient/Laplacian/forcing terms. The present reparametrization keeps $L = 1$ and uses only the U degree of freedom; the achievable spread reduction is still ~ 9 orders of magnitude on the worst cell.

- **The pressure penalty η = eps_val.** The dimensional pressure amplitude P_c varies with the encoding, which means $\eta \cdot p^2$ has a different effective weight per cell. Empirically the discrete LBB stabilization is the primary gauge fix and η acts only as a safety net; the rerun will validate this in practice. If the sweep reveals an η-related regression, a regime-aware scaling $\eta_{\mathrm{cell}} \propto P_c^{-2}$ is a focused follow-up.

- **Solver tolerances.** `ftol`, `xtol`, `stagnation_noise_floor` are absolute. Under the centered encoding the residual magnitudes shrink for low-Re cells; the existing `dynamic_ftol_spatial_safety_factor` machinery scales `ftol` with h^{k+1}, which empirically reaches sensible values in both encodings (the probes converged to high precision). If a regression appears, the dynamic-tolerance formula is the first place to look.

Status

This document accompanies a single-line code change at `test/extended/ManufacturedSolutions/run_test.jl:5` It is kept in `theory/` as a standalone working note pending the full 24-cell sweep rerun on this encoding; once the encoding is shown to be *uniformly* desirable (no regression on the easy cells, demonstrable gains on the high-σ cells), the content here should be folded into `article.tex` as a remark in the verification section.